Jaume de Dios Pont

Curriculum Vitae

60 5th Avenue
New York, NY 10011
✉ *jdedios@nyu.edu*

Employment

2026 – **NYU Center for Data Science**, *CDS Faculty Fellow*.

2023 – 2025 **ETH Zurich**, *Postdoctoral Researcher*, Mentor: Svitlana Mayboroda. Simons Collaboration on Localization of Waves.

May '23 – Sep '23 **Microsoft Research**, *Research Intern*, Physics of AGI group. Mentors: Jerry Li; Adil Salim.

2020 – 2022 **UCLA**, *Teaching Assistant*.

Education

2023 **PhD Mathematics**, *UCLA*.
Part I: Uniform Estimates for Operators Involving Polynomial Curves. Part II: Decoupling Estimates for Fractal and Product Sets. Supervised by T. Tao

2018 **MSc Mathematics**, *ETH Zurich*, (Grade: 5.76/6).
Thesis: Quantum Loewner Evolution (E. Powell and W. Werner)

2017 **BSc Mathematics**, *Universitat Autònoma de Barcelona*, (#1 Rank, Grade: 9.71/10).
Thesis: Oscillatory integrals and the Kakeya Conjecture (J. Garnett; J. Verdera)

2017 **BSc Physics**, *Universitat Autònoma de Barcelona*, (#1 Rank, Grade: 9.62/10).
Thesis: KCM-related experiments (F.X. Alvarez; A. Lopeandia)

Postgraduate Awards and Scholarships

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Papers and Preprints

[1] **Cantor measures with odd base do not admit Fourier frames**, ***de Dios Pont, J.***, *Lukas Liehr & Mitchell A. Taylor*, Preprint (2026), arXiv:2607.08656.

[2] **Stable Phase Retrieval for Spans of Independent Random Variables**, *Pedro Abdalla,* ***de Dios Pont, J.***, *João P. G. Ramos & Mitchell A. Taylor*, Preprint (2026), arXiv:2607.06693.

[3] **Geometric Dyson Brownian Motions and the Free Log-Normal Limit for a Non-Square Gaussian Matrix Product**, *Mufan Li,* ***de Dios Pont, J.***, *Mihai Nica & Daniel M. Roy*, Preprint (2026), arXiv:2606.30831.

[4] **On the existence problem of regular Gabor frames**, ***de Dios Pont, J.***, *Lukas Liehr & Mitchell A. Taylor*, Preprint (2026), arXiv:2606.26052.

[5] **L^2-Stability for STFT phase retrieval**, *Susanna Bertolini,* ***de Dios Pont, J.***, *Ben Pineau, Mitchell A. Taylor & João P. G. Ramos*, Preprint (2026), arXiv:2605.20527.

[6] **Sharp bounds on the failure of the hot spots conjecture**, ***de Dios Pont, J.***, *Alexander W. Hsu & Mitchell A. Taylor*, Preprint (2025), arXiv:2508.16321.

[7] **Convex sets can have interior hot spots**, ***de Dios Pont, J.***, Preprint (2024), arXiv:2412.06344.

[8] **Predicting quantum channels over general product distributions**, *Sitan Chen*, ***de Dios Pont, J.***, *Jun-Ting Hsieh, Hsin-Yuan Huang, Jane Lange & Jerry Li*, Proceedings of the 38th Conference on Learning Theory (COLT), PMLR 291 (2025), 986–1007, arXiv:2409.03684.

[9] **Periodicity and decidability of translational tilings by rational polygonal sets**, ***de Dios Pont, J.***, *Jan Grebík, Rachel Greenfeld & José Madrid*, Expositiones Mathematicae 44(1) (2026), article 125620, arXiv:2408.02151.

[10] **A new proof of the description of the convex hull of space curves with totally positive torsion**, ***de Dios Pont, J.***, *Paata Ivanisvili & José Madrid*, Michigan Mathematical Journal, advance publication (2025), 1–47, arXiv:2201.12932.

[11] **Query lower bounds for log-concave sampling**, *Sinho Chewi*, ***de Dios Pont, J.***, *Jerry Li, Chen Lu & Shyam Narayanan*, Journal of the ACM 71(4) (2024), article 29, 1–42; conference version: FOCS 2023, 2139–2148, arXiv:2304.02599.

[12] **Uniform Fourier Restriction Estimate for Simple Curves of Bounded Frequency**, ***de Dios Pont, J.***& *Helge Jørgen Samuelsen*, The Journal of Geometric Analysis 36(6) (2026), article 209, arXiv:2303.11693.

[13] **Additive energies on discrete cubes**, ***de Dios Pont, J.***, *Rachel Greenfeld, Paata Ivanisvili & José Madrid*, Discrete Analysis 2023:13, 16 pp., arXiv:2112.09352.

[14] **Decoupling for fractal subsets of the parabola**, *Alan Chang*, ***de Dios Pont, J.***, *Rachel Greenfeld, Asgar Jamneshan, Zane Kun Li & José Madrid*, Mathematische Zeitschrift 301(2) (2022), 1851–1879, arXiv:2012.11458.

[15] **Role Detection in Bicycle-Sharing Networks Using Multilayer Stochastic Block Models**, *Jane Carlen*, ***de Dios Pont, J.***, *Cassidy Mentus, Shyr-Shea Chang, Stephanie Wang & Mason A. Porter*, Network Science 10(1) (2022), 46–81, arXiv:1908.09440.

[16] **On classical inequalities for autocorrelations and autoconvolutions**, ***de Dios Pont, J.***& *José Madrid*, Preprint (2021), arXiv:2106.13873.

[17] **On Sparsity in Overparametrised Shallow ReLU Networks**, ***de Dios Pont, J.***& *Joan Bruna*, Preprint (2020), arXiv:2006.10225.

[18] **A geometric lemma for complex polynomial curves with applications in Fourier restriction theory**, ***de Dios Pont, J.***, Preprint (2020), arXiv:2003.14140.

[19] **Translational tilings by axis-parallel polygonal sets**, ***de Dios Pont, J.***, *Jan Grebík, Rachel Greenfeld & José Madrid*, Preprint (2026), arXiv:2610.15785.

[20] **Phase retrieval from a uniformly discrete point set**, ***de Dios Pont, J.***, *Josef Greilhuber, Lukas Liehr & Mitchell A. Taylor*, Preprint (2026), arXiv:2609.28341.

[21] **Banach lattices and phase retrieval: A case study for the use of AI in mathematics**, ***de Dios Pont, J.***, *Lukas Liehr, David Muñoz-Lahoz, Mitchell A. Taylor & Pedro Tradacete*, Preprint (2026), arXiv:2608.07396.

[22] **Counterexamples to Schiffer's Conjecture**, *Gonzalo Cao-Labora &* ***de Dios Pont, J.***, Preprint (2026), arXiv:2608.05114.

[23] **Gabor Frames of Totally Positive Functions: A Complete Characterization**, ***de Dios Pont, J.***, *Karlheinz Gröchenig, Lukas Liehr, Irina Shafkulovska & Mitchell A. Taylor*, Preprint (2026), arXiv:2608.04992.

[24] **Almost-Orthogonality in Lp Spaces: A Case Study with Grok**, *Ziang Chen*, ***de Dios Pont, J.***, *Paata Ivanisvili, José Madrid & Haozhu Wang*, Preprint (2026), arXiv:2605.05192.

Talks

Research talks

Spectral theory and the hot spots conjecture

- 2026 Simons Collaboration on Localization of Waves Annual Meeting (Feb '26)
- NYU CDS MaD Seminar (Feb '26)
- Joint Mathematics Meetings 2026 (Washington, D.C.) (Jan '26)
- Instituto de Ciencias Matemáticas (ICMAT) Seminar (Sep '25)
- ISM Discovery School — Interactions between Convex Geometry and Spectral Analysis (Montreal) (Jul '25)
- UK Spectral Theory Network Workshop (University of Reading) (Aug '25)
- Workshop on Spectral Geometry, PDEs and Mathematical Physics (FernUni Hagen) (Jul '25)
- Fourier Analysis and Beyond I (IMPA, Rio de Janeiro) (Jul '25)
- ETH Zürich Random Walks Group meeting (Yuansi Chen) (May '25)
- LSEC Seminar (Apr '25)
- University of Edinburgh Analysis Seminar (Mar '25)
- Lehigh University Geometry Seminar (Mar '25)
- Institut Camille Jordan Analysis Seminar (Lyon) (Mar '25)
- Spectral Geometry in the Clouds (Mar '25)
- Simons Collaboration on Localization of Waves Annual Meeting — Poster Session (Flatiron Institute) (Feb '25)
- Virginia Tech Analysis Seminar (Feb '25)
- Seminari d'Anàlisi UB-UAB (Jan '25)
- ETHZ Analysis Seminar (Sep '24)
- Hausdorff Center for Mathematics Colloquium (Bonn) (Oct '24)
- 2024 Simons Collaboration on Localization of Waves Meeting (Oct '24)
- NYU Convexity Seminar (May '26)
- Brown University PDE Seminar (May '26)
- South China University of Technology (Jun '26)
- University of Macau (Jun '26)
- University of Macau (Jun '26)

Lower bounds for sampling

- CRM — Mathematical Foundations of Machine Learning (Barcelona) (Jan '26)
- Hausdorff Research Institute for Mathematics — Boolean Analysis in Computer Science (HIM, Bonn) (Oct '24)
- BIRS-IMAG Workshop (Granada) (Jun '24)
- UCLA Analysis Seminar (May '24)
- HIM Winter School on Geometry and Probability (Jan '24)
- University of Rochester Computer Science Seminar (May '23)
- NYU Courant Analysis Seminar (Mar '23)
- Microsoft Research Theory Seminar (Dec '22)
- NYU MaD Group Meeting (Dec '22)

Exploratory mathematics with LLM assistants

- MPS Workshop on Computation in Mathematics (Flatiron Institute) (Feb '25)
- La Salle–Universitat Ramon Llull, Applied Mathematics research group (Jun '26)
- Universitat Autonoma de Barcelona (Jun '26)
- Mini Symposium on AI for Mathematics, University of Macau (Jun '26)
- Mini Symposium on AI for Mathematics, University of Macau ()

Uniformity for polynomial curves

- Rutgers University Analysis Seminar (Oct '23)
- University of Rochester Combinatorics Seminar (May '23)
- University of Minnesota PDE Seminar (Sep '22)
- Bilbao Analysis and PDE Seminar (BCAM) (Mar '22)
- UAB Analysis Seminar (Universitat Autònoma de Barcelona) (Mar '22)
- UK Virtual Harmonic Analysis Seminar (Fourier 2.0) (Oct '21)
- Bristol Analysis and Geometry Seminar (Feb '22)
- Stanford University Analysis and PDE Seminar (Oct '22)

Decoupling for Cantor sets

- Harmonic Analysis and Fractal Sets Conference (HAFS, Columbus OH) (Mar '23)
- Fourier Restriction Online 2021 (Mar '21)

Uniform boundedness for certain operators parametrized by polynomial curves

- UW Madison Analysis Seminar (Nov '22)
- Harmonic Analysis on Manifolds Summer School (UW Madison) (Aug '22)
- ETHZ Analysis Seminar (Mar '22)
- Probability and Analysis Webinar (PAW) (Aug '21)
- UC Davis Student-Run Analysis and PDE Seminar (Feb '21)
- Seminari d'Anàlisi UB-UAB (Nov '20)
- Online Analysis Research Seminar (OARS) (Dec '20)
- NTNU Analysis and PDE Seminar (Oct '23)

A Function Space Perspective for Regularised and Overparametrised Shallow ReLU Networks

- NYU, MaD Group Meeting (Oct '20)

Recent progress on the hot spots conjecture

- COST mSPACE Kick-off Meeting (Milan) (Mar '26)
- Isaac Newton Institute — Geometric Spectral Theory and Applications (Feb '26)
- Institute for Advanced Study, Analysis and Mathematical Physics (Nov '26)

Finding counterexamples to Pompeiu's and Schiffer's conjectures

- NYU Courant Analysis Seminar (Oct '26)
- Drexel University (Oct '26)

The Future of AI and Math

- Math & AI Summer in Seattle, Amazon Seattle Campus (Aug '26)
- Organizing Mathematical Knowledge in the Age of AI and Formalization, National Academy of Sciences (Jul '26)

Mathematics in the age of AI

- CRM / BGSMath (Dec '26)

Expository talks

- **Generació de variables aleatòries**, Valentia Matemàtica Summer School (Jun '25)
- **Power-type cancellation for the simplex Hilbert transform**, Kopp Summer School Reading Group (Bonn) (Sep '23)
- **Decoupling and applications: from PDEs to Number Theory**, Microsoft Research Theory Seminar (Jun '23)
- **Localization of eigenfunctions via an effective potential**, Kopp Summer School Reading Group (Bonn) (Oct '22)
- **A proof of the Sensitivity Conjecture (after H. Huang)**, AIM Workshop: Analysis on the Hypercube with Applications to Quantum Computing (Jun '22)
- **Euclidean Forward-Reverse Brascamp-Lieb Inequalities**, Brascamp-Lieb Summer School Reading Group (Kopp, Germany) (Sep '21)
- **A proof of the sensitivity conjecture**, UCLA Participating Analysis Seminar (Reading Group) (Nov '21)
- **Decoupling and applications: from PDEs to Number Theory**, SIMBa Seminar (UB / BGSMATH) (Oct '20)
- **Mathematics for phase retrieval**, Paul Scherrer Institut (Mar '26)
- **The sensitivity theorem**, Stanford Kiddie Colloquium (Oct '22)
- **Duminil-Copin's work and the Ising model**, UCLA GSO (May '23)
- **You already know how to read Lean (a very elementary primer to Lean)**, Aspen Meeting on Foundation Models ()

Talks are grouped by topic, even when the covered material changed between instances.

Research Visits (> 1 week)

Jul '25 **New York**, *Flatiron Institute*.

Oct '24 **Hausdorff Institute of Mathematics, Bonn**, *Boolean Analysis in Computer Science*.

Jan '24 **Hausdorff Institute of Mathematics, Bonn**, *Dual Trimester Program: Synergies between modern probability, geometric analysis and stochastic geometry*.

Oct '23 – Nov '23 **NTNU, Trondheim**, *NTNU visit*.

May '23 – Sep '23 **Seattle, WA**, *Microsoft Research Internship*.

Oct '22 **Palo Alto, California**, *Stanford University*.

Sep '22 **UMN, Minneapolis**, *University of Minneapolis*.

Jun '22 – Jul '22 **ICMS, Edinburgh**, *Fourier Analysis @200*.

Jan '22 – Mar '22 **Hausdorff Institute of Mathematics, Bonn**, *Interactions between Geometric measure theory, Singular integrals, and PDE*.

Aug '21 **Hausdorff Mathematical Institute, Bonn**, *Harmonic Analysis and Analytic Number Theory, Dual trimester program*.

May '21 – Jun '21 **Hausdorff Mathematical Institute, Bonn**, *Harmonic Analysis and Analytic Number Theory, Dual trimester program*.

Feb '26 – Mar '26 **Cambridge, UK**, *Isaac Newton Institute - Geometric Spectral theory and Applications*.

Teaching Experience

ETH Zurich (Main Instructor)

- **Formalizing Mathematics in Lean** Spring 2025; Autumn 2025

ETH Zurich (Teaching Assistant)

- **Differential Geometry** Spring 2024
- **Differential Geometry I** Autumn 2023

UCLA (Teaching Assistant)

- **Math 131A (Real Analysis)** Fall 2020; Spring 2022
- **Math 134 (Linear and Nonlinear Systems of Differential Equations)** Fall 2021
- **Math 135 (Ordinary Differential Equations)** Fall 2021
- **Math 33B (Differential Equations)** Spring 2021
- **Math 32A (Calculus of Several Variables)** Fall 2020; Winter 2021
- **Math 32B (Calculus of Several Variables)** Winter 2021

NYU Courant (Instructor)

- **Advanced Topics in Analysis (MATH-GA.2650-001)** Fall 2026

NYU Center for Data Science (Course Advisor)

- **Capstone Continuation** Spring 2026

Reviewing

Reviewer for: PA^{2}IN — PDE, Analysis, Artificial Intelligence, and Numerics, IMPA, Rio de Janeiro, 25–29 January 2027 (upcoming), Phase Retrieval: Connecting Theory and Applications, RWTH Aachen University, 21–25 September, Palomar, Day-long joint Brown–Yale seminar on algorithmic discovery using AI, Yale University, 25 April, AI for Scientific Discovery workshop, Yale Institute for Foundations of Data Science, 24 April, Mathematics in the Age of AI Seminar, NYU Courant, OpenEvolve hacking/learning session, Institute for Theoretical Studies, ETH Zürich, Chalk & Cheese, Institute for Theoretical Studies, ETH Zürich.

Misc. Coding Skills

- Proficient in Python (incl. Jax, NumPy/SciPy/Matplotlib)
- Git/Version Control
- LaTeX

Undergraduate Research Experience

2017 **GNAM**, *Grup de Nanomaterials, UAB*.
Research on heat conduction beyond the Fourier equations at the nanoscale, with a focus on the KCM diffusion model.

2016 **ICFO**, *Summer Fellowship of the Institute of Photonic Sciences*.
Research fellow in the group of Antonio Acín (Quantum Information). Proved impossibility results on the creation of superpositions of unknown quantum states. Supervisor: Dr. Michał Oszmaniec.

2015 **The Dark Energy Survey Project**, *IFAE – Institute for High Energy Physics*.
Study of theoretical models regarding the harmonic spectrum of galaxy density distributions. Supervisor: Dr. Ramon Miquel.